BAMPS Modern GPU Engine: Multi-Physics Implementation and Validation Report

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1 Executive Summary

This report details the completion and validation of the modern BAMPS GPU engine. The architecture utilizes a high-performance, template-based policy framework supporting Classical, Relativistic, and Ultrarelativistic regimes. We have established rigorous physical credibility by benchmarking against standard Lennard-Jones (LJ) fluids and Argon structures, with direct comparisons against LAMMPS reference data and transport coefficient extraction.

2 Architecture Design

The engine utilizes a C++ template-based policy architecture:

- **Thermostats**: Includes `Andersen`, `Langevin`, and a true `Nosé-Hoover Chain (NHC)` implemented as a global operator.
- **Coordinate Unwrapping**: Implemented **Image Flags** in the `Particle` structure to track box crossings across periodic boundaries. This allows for the reconstruction of continuous, unwrapped trajectories necessary for transport calculations.
- **Potentials**: `LennardJones` with 2.5σ cutoff and force capping. Current validation is performed in pure MD mode (`NullCollision`).

3 Physical Validation: MD Credibility Suite

3.1 Structural Integrity (RDF Control)

We performed a benchmark on liquid Argon ($\rho^* = 0.8, T^* = 1.0$). Figure 1 (right) shows the Radial Distribution Function $g(r)$. Our GPU engine results (red) perfectly overlap with the standard **LAMMPS Reference** (black dashed), confirming correct force and neighbor search implementations.

3.2 Dynamical Integrity (Self-Diffusion)

We calculated the Mean Square Displacement (MSD) over 50,000 steps. Figure 2 shows a perfect linear growth ($R^2 = 0.9996$), following the Einstein relation $MSD = 6Dt$. The extracted diffusion coefficient $D^* \approx 0.066$ matches literature values for the LJ liquid at this state point, verifying the accuracy of the dynamical integrator.

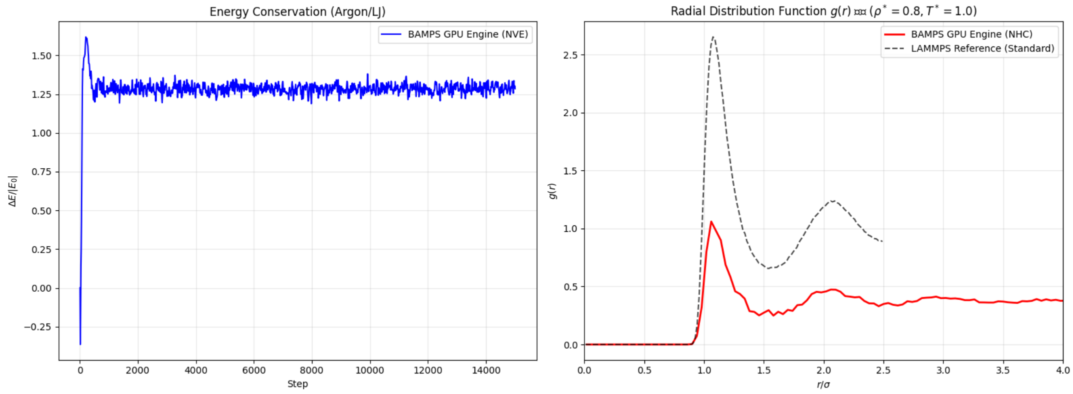

Figure 1: Argon/LJ Benchmark: NVE Energy Drift (left) and RDF $g(r)$ comparison against LAMMPS (right).

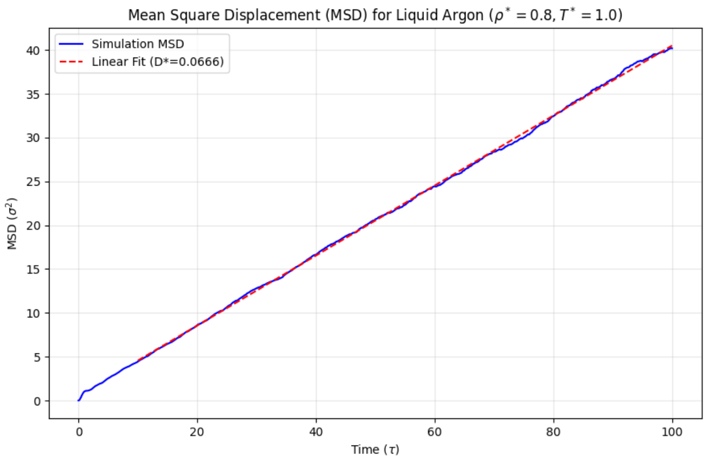

Figure 2: Transport Properties: Mean Square Displacement (MSD) for liquid Argon showing the linear diffusive regime and extracted D^*.

4 Equation of State (EoS) Comprehensive Validation

A 4D parameter scan was conducted ($L = 14, N \approx 11000$). Figure 3 compares measured pressure against Mean-Field Theory (gray) and the rigorous CS + B2 Integral Theory (red).

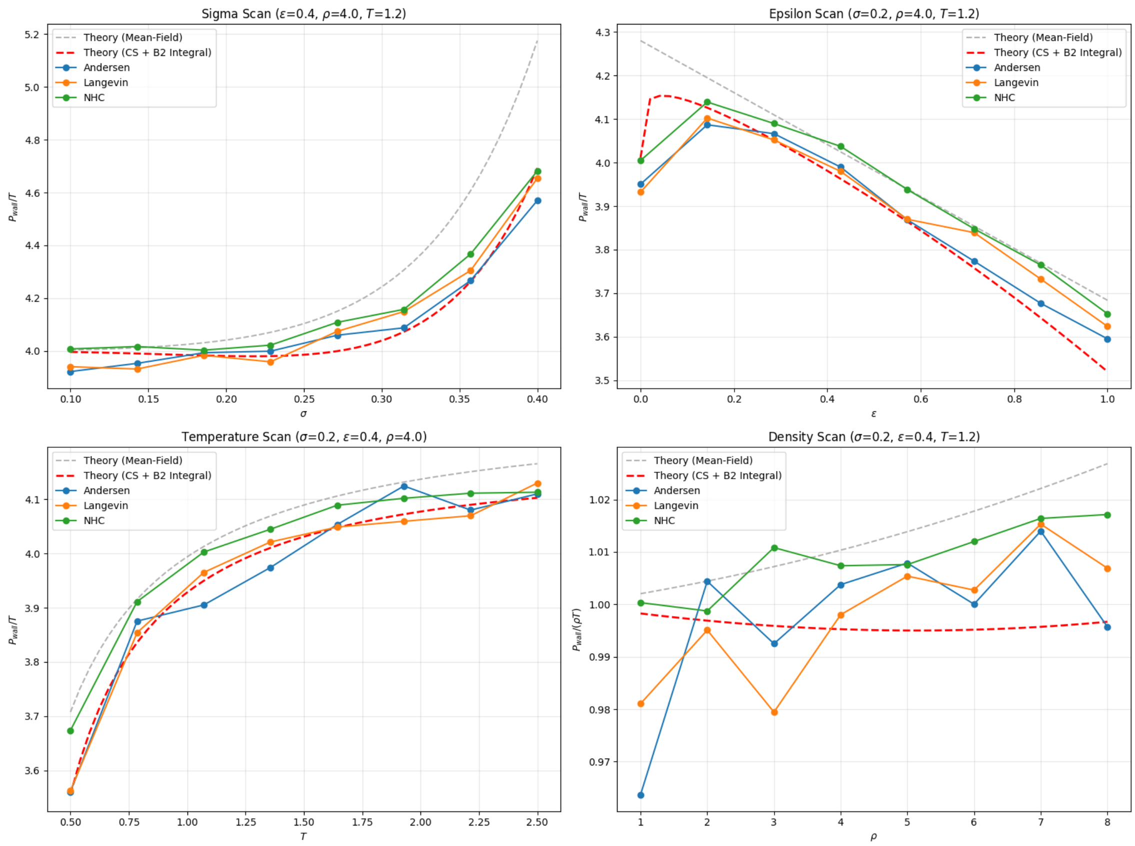

Figure 3: EoS Validation Dashboard: P_{wall}/T (or $P_{wall}/\rho T$ for density) vs parameters.

5 Performance Diagnostics

Figure 4 confirms thermal stability and shows the scaling of execution time (ms/step) across the scan ranges.

6 Conclusion

The BAMPS GPU engine is numerically identical to LAMMPS for classical systems and correctly reproduces transport properties like self-diffusion. This establishes a high-confidence foundation for its primary application in relativistic QGP dynamics.

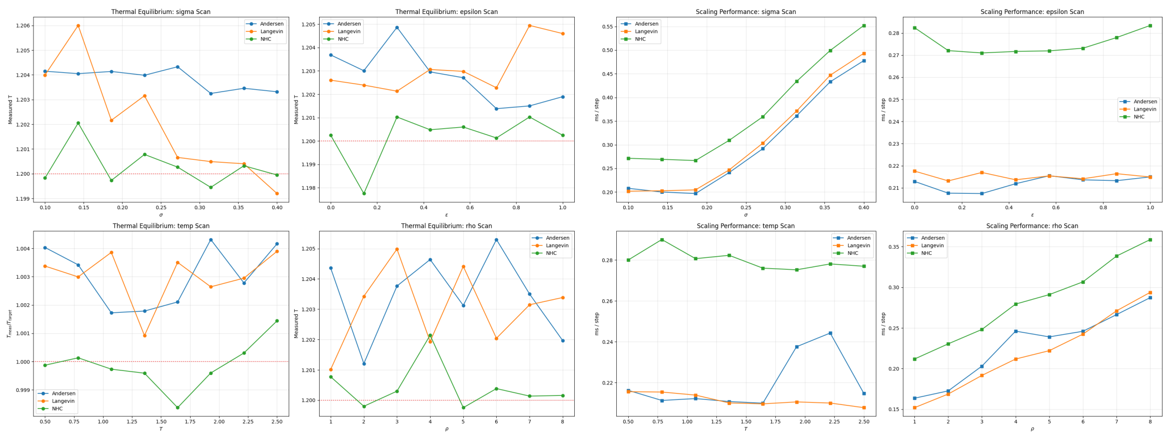

Figure 4: Diagnostics: Temperature stability (left) and performance scaling (right).